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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/703,736	04/23/2024	Hitoshi SUMIYA	SUMI29PUS01	5657

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EXAMINER
CULLER, JILL E

ART UNIT	PAPER NUMBER
2853	

NOTIFICATION DATE	DELIVERY MODE
04/07/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

ipdocket@rennerotto.com

Office Action Summary	Application No. 18/703,736	Applicant(s) SUMIYA et al.	
	Examiner Jill E Culler	Art Unit 2853	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on April 23, 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-7 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on April 23, 2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to

file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

Claims 1-7 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-7 of copending Application No. 18/707,428 (reference application). Although the claims at issue are not identical, they are not patentably distinct from each other because they teach the same general subject matter.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

With respect to claim 1, Claim 1 of 18/707,428 teaches all that is claimed except for claiming different size ranges for the length from the rotation axis to the outer end portion and the average particle diameter of the plurality of diamond particles. Because the changes in these parameters do not significantly change the mode of operation of the contactor, it would

have been obvious to one having ordinary skill in the art at the time the invention was filed to choose an optimum value for these parameters through routine experimentation and accordingly the differences do not patentably distinguish the two inventions.

With respect to claim 2, Claim 2 of 18/707,428 teaches all that is claimed except for claiming a different value of Knoop hardness. As with claim 1, because the changes in these parameters do not significantly change the mode of operation of the contactor, it would have been obvious to one having ordinary skill in the art at the time the invention was filed to choose an optimum value for these parameters through routine experimentation and accordingly the differences do not patentably distinguish the two inventions.

With respect to claim 3, Claim 3 of 18/707,428 teaches all that is claimed.

With respect to claim 4, Claim 4 of 18/707,428 teaches all that is claimed.

With respect to claim 5, Claim 5 of 18/707,428 teaches all that is claimed.

With respect to claim 6, Claim 6 of 18/707,428 teaches all that is claimed.

With respect to claim 7. Claim 7 of 18/707,428 teaches all that is claimed.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having

ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) 1-3 is/are rejected under 35 U.S.C. 103 as being unpatentable over Fukunishi et al. (CN 102729349, hereafter Fukunishi)

With respect to claim 1, Fukunishi teaches a contactor (scribing wheel 50) having an annular shape and being made of polycrystalline diamond including a plurality of diamond particles, the contactor (scribing wheel 50) being configured to have a rotation axis (50b) extending through a center of the contactor, the contactor comprising: a first portion (body portion 51) including an inner end portion, the first portion having a constant thickness in a radial direction; and a second portion (cutter 52) including an outer end portion, the second portion having a decreasing thickness in the radial direction, wherein the second portion has a first surface continuous with an upper surface of the first portion, a second surface continuous with a lower surface of the first portion, and a connection surface connecting the first surface and the second surface to each other and including the outer end portion, wherein in a cross-section of the contactor along the rotation axis, an angle θ (angle θ2) formed by a first line segment indicating the first surface and a second line segment indicating the second surface is 100º to 150º, and a length from the rotation axis to the outer end portion is 0.5 mm to 5 mm. (Abstract, trans. pars. 65-68, 87-100, Figs. 3, 7)

Fukunishi does not explicitly teach a length between a first boundary portion and a second boundary portion is 1 μm to 10 μm, the first boundary portion being a boundary between the first surface and the connection surface, the second boundary portion being a boundary between the second surface and the connection surface or wherein the plurality of diamond particles have an average particle diameter of 10 nm to 300 nm.

However, the optimum length between the boundary portions and the optimum particle diameter are both variables which depend upon the other components of the system and the intended use of the object and therefore both could have best been determined through routine experimentation.

With respect to claim 2, although Fukunishi does not explicitly discuss a hardness value for the polycrystalline diamond, it would have been obvious to one having ordinary skill in the art at the time the invention was filed to improve the hardness to a high level in order to provide a strong and durable contacting surface.

With respect to claim 3, Fukunishi teaches a line of intersection of the connection surface and a cross-section including the rotation axis is a straight line. (Figs. 3, 7)

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. US 6,361,412, WO 2015022873; CN 108608588; US 10,173,316 and CN 106313346 each teach a contactor having similarities to the claimed subject matter.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Jill E Culler whose telephone number is (571)272-2159. The examiner can normally be reached M-F 8:30-5:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is

encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Stephen Meier can be reached at 571-272-2149. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JILL E CULLER/
Primary Examiner, Art Unit 2853